

Part II Applied (2 hours) December 5th, 2015

1) A researcher is studying the changes from 1850 to 1950 in lead in hair of children and adults using two independent samples. The studies only gave the summary numbers below, and it was impossible to get the original raw data. The response variable Y is the amount of lead in milligrams per gram of hair is approximately normally distributed, and has homogeneous variance.

- test whether there has been a change from 1850 to 1950 in lead exposure in the population as a whole
- test whether adults have higher exposure than children
- summarize your results in an appropriate table. In addition summarize in a few sentences what the data shows.

PARAMETER	ESTIMATE	STD-ERROR
a	40	1.1
b ₁	-0.1	0.025
b ₂	-2	0.5
b ₁₂	-0.5	0.025

A. Give a 95% confidence interval for σ^2 or if you cannot answer the question from the information given explain what additional information you would need to answer the question

B. Given a 95% confidence interval for the mean hemoglobin of a 30 year old woman or if you cannot answer the question from the information given explain what additional information you would need to answer the question

- 3) Dr. Lee has developed The “Dooku-Test” (DT) for infection by the Dooku virus.
- I. He tests 1,000 persons who truly have the Dooku virus with the DT and
 - correctly identifies that 990 have the Dooku virus
 - falsely identifies that 10 do not have the Dooku virus
 - II. He then tests 1,000 persons who truly don’t have the Dooku virus with the DT and
 - correctly identifies that 990 do not have the Dooku virus
 - falsely identifies that 10 do have the Dooku virus

Remember that the numbers in I and II are sample estimates. Suppose in the United States that exactly 1/1000 of the persons (0.001) are infected with the Dooku virus (which means 0.999 are not). You should take this number to be known rather than an estimate

- A. Give the best point estimates for the:
 - a. Probability that a US person is both infected with the Dooku virus and would test positive for Dooku by the DT.
 - b. Probability that a US person is both not infected with the Dooku virus and would test positive for Dooku by the DT.
 - B. Give a large sample 95% confidence interval for the point estimate in A-a. up above (Do this only for the estimate in A-a, not for the estimate in A-b)
 - C. If a randomly chosen US person tests positive for the Dooku virus by the DT, give the best point estimate for the probability they are really infected with that virus.
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PROBLEM 4 IS ON THE NEXT PAGE

4) A researcher is studying the effect of two medicines on reducing Cholesterol; 30 individuals are randomly divided into 3 groups of 10 each. The first group gets Medication A, the second gets Medication B, and the third gets a placebo. Cholesterol levels (in mg/dl) are measured for each individual at two times i) before taking the medicine/placebo and ii) after taking the medicine/placebo. The response variate Y is reduction in Cholesterol level (Before level – After level). The ages of individuals ranged from 30 to 70, with an overall average of 47.87 years. The following least-squares model is fit to the data: (The errors are normally distributed and independent of the variables).

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 \text{Age} + \beta_4 X_1 * \text{Age} + \beta_5 X_2 * \text{Age} \quad (1)$$

Where medication groups are coded as follows:

	X1	X2
Placebo	-1	-1
Medicine A	1	0
Medicine B	0	1

The results on the fit of the full model (1) are:

Sum of Squares of Error = 4,117; Sum of Squares of model = 20,457

Term	Estimate	Standard Error
Intercept	64.75	10.31
X ₁	41.14	3.41
X ₂	65.82	3.41
Age	1.73	0.21
(Age - 47.87)*X ₁	0.40	0.32
(Age - 47.87)*X ₂	1.61	0.28

a) Use the full model 1 to estimate the average Y for 60 year olds using Medicine A.

b) for the full model 1, test (at $\alpha = 0.05$) which terms are statistically significant.

c) The model without the interaction terms is

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 \text{Age} \quad (2)$$

The Sums of Squares of error for model (2) is 14,551. Using this information together with the corresponding information for model (1), test whether we can drop the two interaction effects, and just use model 2